Allocator confound study: PR #2455 against its merge-base, 2026-10-23

Study `research/ecbench_allocator_confound_20261023`, preregistered in `PREREGISTRATION.md` (commit 6276977d5) before any data, amended once in `AMENDMENT_1.md` (exploratory arms, written before their data). Observations only: wall time is descriptive on a shared host and never the headline (AGENTS.md section 6). Every number below is a recorded value or a plain order statistic of recorded values (jq, `analysis/summarise.sh`), or an `ecbench compare`, `band` or `verify` output; Python only drew the graphs.

Headline

- **Counts are identical (exact comparison).** For 10 base-versus-variant session pairs every count-bearing field of every record matches: 72 of 72 (smoke) and 66 of 66 (claim) records per pair, 0 differences, 0 unmatched. All 24 per-arm `ecbench compare --b-dir` read-outs give ops ratio 1.0, interval [1, 1], `same_seeds` true. A tampered copy is flagged by the same script (negative control).
- **The PR binary reads a much shorter solve wall on the small-set arms (measured, descriptive).** smoke-prime, 8 rho/BSGS/kangaroo arms, n = 12 per arm per binary: pooled median 5.26 to 6.77 ms on the merge-base binary, 0.137 to 0.330 ms on the PR binary, ratio 18.4 to 38.8; in every arm all 12 PR-binary runs lie below the merge-base median. The index-calculus arm `ic-mitm` (about 1.9 ms) does not move (ratio 1.03, 58% of PR runs below).
- **The IC online window on claim-koblitz is not distinguished (measured, descriptive).** n = 40 per binary: median 127.1 us on the merge-base binary against 119.1 us on the PR binary (ratio 1.067, 65% of PR runs below the base median); by order replicate 1.135 and 0.983. The PR text's "about 15% lower" is not reproduced; the preregistered verdict is inconclusive. The whole solve of `rho` and `rho-strong` on claim-koblitz is 23.6 and 38.0 times shorter, as in smoke.
- **Attribution, tested by scratch builds of the PR source (measured, exploratory).** Removing only `malloc_trim(0)` leaves the PR binary just as fast; removing only the new pre-solve `SolveProbe::read` (the `/proc` reads) leaves it just as fast; removing both returns every arm to the merge-base level (ratio 0.75 to 1.05). Either one consolidates the allocator before the clock opens. The trim removes the confound and does so by design; in this build it is not the only thing that does.
- **Audits.** All 14 sessions pass `ecbench verify --replay-all --exit-code` (the PR binary as auditor) and the CI form `--replay 0 --allow-interrupted --exit-code`.
- **Gap to rho is unchanged and is not the point.** On claim-koblitz (Koblitz a=1, n=17) the section 5a band gives IC against the matched `rho-strong` S of 16.85 against 4.00 (w = 0), 17.28 against 4.14 (0.1), 21.16 against 5.36 (1): rho-strong ahead at every price, IC 4.21, 4.17 and 3.95 times it. This study corrects a measuring instrument; it is not an index-calculus improvement.

Design

arm	what it is	binary sha256 (prefix)
`base`	merge-base `3a1ea541d`, `git archive`, own target dir	a59c4014ff52
`fix`	`ecbench/fix-runner` source at `63fede816` (the PR; later branch commits touch no source)	26f25d9fa47a
`fix-notrim`	`fix` with `malloc_trim(0)` replaced by `Some(0)` (`analysis/notrim.patch`)	in `BINARIES.txt`
`fix-noprobe`	`fix` with `SolveProbe::read` reading nothing (`analysis/noprobe.patch`)	in `BINARIES.txt`
`fix-notrim2`	both removed (`analysis/notrim2.patch`)	in `BINARIES.txt`

Specs: committed `docs/ecbench/specs/smoke-prime.json` (20-bit prime-order curve, 9 arms including the A/A control `rho-neg-aa`, 2 targets, 3 rounds + 1 warm-up) and the new `specs/claim-koblitz-r10.json` (the committed `claim-koblitz.json` with rounds 1 to 10: Koblitz a=1 n=17, 2 public targets, arms `rho-strong`, `rho`, `ic`). `base` and `fix` were run in ABBA order (`base`, `fix`, `fix`-rev, `base`-rev) to cancel drift and pooled per binary: smoke n = 12, claim n = 40 per arm. The three ablations ran once per spec after the ABBA sessions (smoke n = 6, claim n = 20). Every session: `ecbench run --wait --allow-busy --cpus auto --quiet`, commands in `run_all.sh` and `logs/`.

Host: one shared 4-vCPU Xeon 2.10 GHz VM (`ecbench host` class ECBENV3had995aa2df0f) with other agents building and testing throughout (load average 2.7 to 5.1 at session starts, the cargo lock held every time; `logs/run_all.log`). No run reached L3. Isolation levels recorded per record:

group	levels (record counts)
`claim-base`	L1x120
`claim-fix`	L1x120
`claim-noprobe`	L0x60
`claim-notrim`	L1x60
`claim-notrim2`	L0x60
`smoke-base`	L1x60 L2x48
`smoke-fix`	L1x102 L2x6
`smoke-noprobe`	L0x54
`smoke-notrim`	L1x48 L2x6
`smoke-notrim2`	L0x54

The ablation sessions with `SolveProbe::read` stubbed carry L0 (blocker recorded: "the child did not report its thread count"), and every PR-source record whose solve is under 1 ms carries the blocker "solve too short to admit wall time" (grading v3), which is why the PR binary's smoke levels differ from the merge-base binary's (graded under v2). Their wall values are recorded and reported as descriptive only.

Results

Counts: identical across binaries

base session	compared session	records matched	differing	unmatched	result
base-smoke	fix-smoke	72	0	0	identical
base-smoke-rev	fix-smoke-rev	72	0	0	identical
base-claim	fix-claim	66	0	0	identical
base-claim-rev	fix-claim-rev	66	0	0	identical
base-smoke	fix-notrim-smoke	72	0	0	identical
base-claim	fix-notrim-claim	66	0	0	identical
base-smoke	fix-notrim2-smoke	72	0	0	identical
base-claim	fix-notrim2-claim	66	0	0	identical
base-smoke	fix-noprobe-smoke	72	0	0	identical
base-claim	fix-noprobe-claim	66	0	0	identical

Fields compared per record (analysis/counts_diff.sh): status, recovered logarithm, matches_target, algorithm_seed, run_id, workload and method hashes, total_gae, S, lower_bound, unpriced, per-phase adds, doubles, scalar multiplications, GAE and native counters, and all counters. ecbench compare per arm (comparisons/, analysis/ops_compare.tsv): ratio 1.0, interval [1, 1], same_seeds true in all 24. analysis/counts_diff_negative_control.log: a copy of fix-smoke with one bsgs record's counters raised by 1 is reported as 1 differing record (exit 1), so the check can fail.

Allocator block (record.allocator): every record of a PR-source binary carries pre_solve_trim: true; base records carry no block.

group	measured records	with pre_solve_trim true	no allocator block
claim-base	120	0	120
claim-fix	120	120	0
claim-noprobe	60	60	0
claim-notrim	60	60	0
claim-notrim2	60	60	0
smoke-base	108	0	108
smoke-fix	108	108	0
smoke-noprobe	54	54	0
smoke-notrim	54	54	0
smoke-notrim2	54	54	0

Smoke: solve wall by arm

arm	n base	n fix	base median ms	fix median ms	base / fix	share of fix runs below base median	ns per GAE base	ns per GAE fix
rho-neg	12	12	5.833	0.25	23.37	100%	6767	291
rho-plain	12	12	6.144	0.277	22.22	100%	5215	245
rho-frozen	12	12	6.091	0.33	18.43	100%	3805	186
rho-neg-aa	12	12	6.767	0.25	27.03	100%	7976	267
bsgs	12	12	5.392	0.186	28.98	100%	4975	168
bsgs-il	12	12	5.262	0.162	32.52	100%	10222	267
bsgs-neg	12	12	5.303	0.137	38.76	100%	7831	186
kangaroo	12	12	6.24	0.196	31.8	100%	4458	123
ic-mitm	12	12	1.891	1.83	1.03	58%	247	221

Wall values are time.solve_wall_ns of verified measured records, pooled over the two orders; ns per GAE is the median of the per-record solve_wall_ns / total_gae. The A/A pair rho-neg and rho-neg-aa (same method) differ by a factor 1.160 on the merge-base binary and 1.003 on the PR binary (table below): the arm-dependent bias the PR describes shows up as an A/A disagreement before and not after, descriptively. The grader of the PR binary (v3) refuses to admit wall time from a solve under 1 ms, so the PR-binary smoke figures are recorded but not admitted wall; no ecbench admitted-wall claim is made.

group	rho-neg median ms	rho-neg-aa median ms	aa / neg
smoke-base	5.833	6.767	1.16
smoke-fix	0.25	0.25	1.003
smoke-notrim	0.236	0.221	0.936
smoke-noprobe	0.277	0.278	1.005
smoke-notrim2	6.57	6.419	0.977

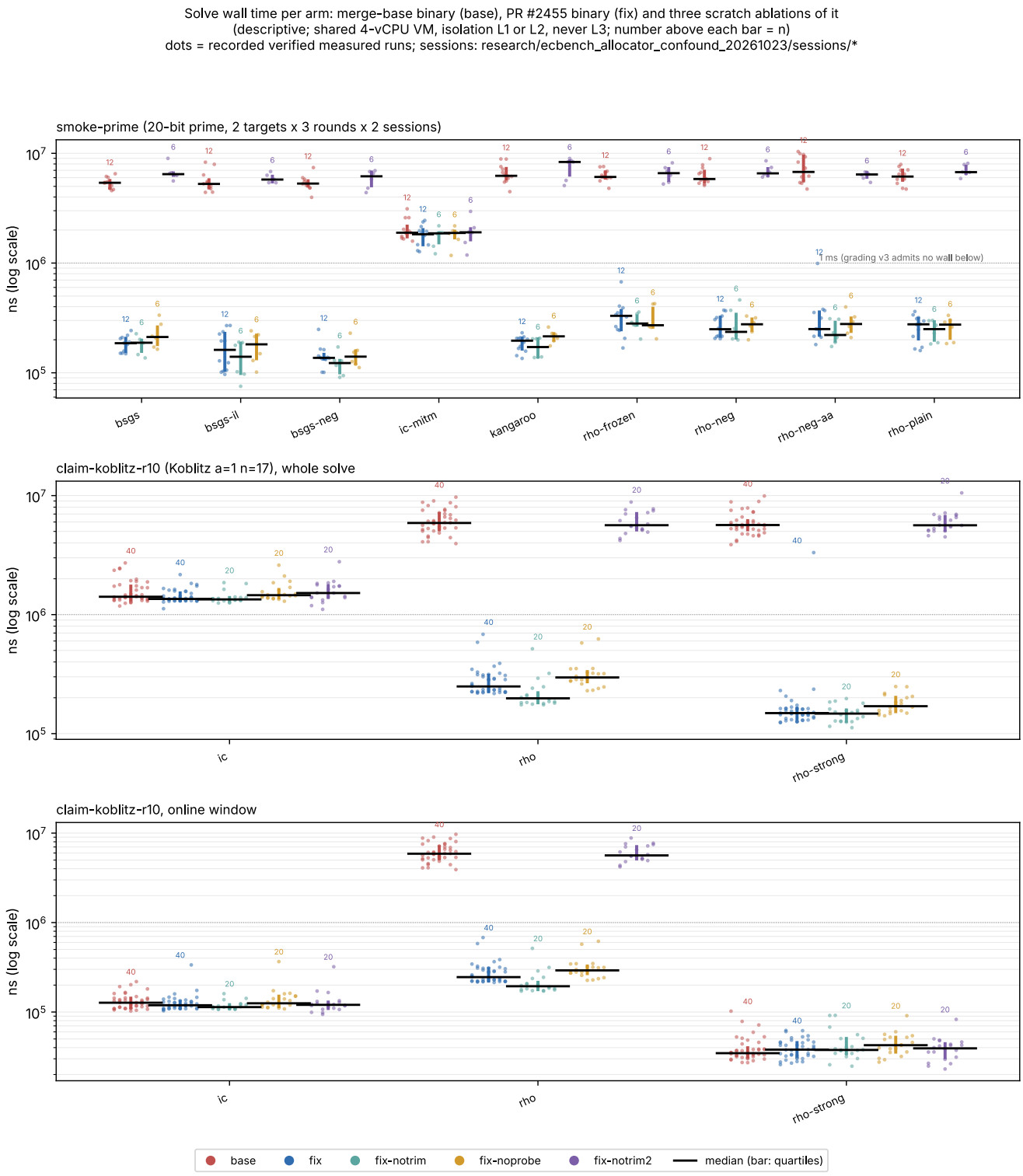


Figure 1: Recorded solve wall time per arm (dots, log scale), pooled median and quartile bar, n above each bar; merge-base binary, PR binary and three scratch ablations. Sessions in `sessions/`; data in `analysis/summary.json`. Descriptive wall, shared host, L1 or L2.

claim-koblitz: solve and online window

arm	window	n base	n fix	base median us	fix median us	base / fix pooled	order 1	order 2 (rev)	share of fix runs below base median
`ic`	solve	40	40	1412.7	1354.2	1.043	1.165	1.025	68%
`ic`	online	40	40	127.1	119.1	1.067	1.135	0.983	65%
`rho`	solve	40	40	5886.9	249.1	23.637	25.125	21.967	100%
`rho`	online	40	40	5885.3	245.6	23.965	25.462	22.283	100%
`rho-strong`	solve	40	40	5658.4	148.7	38.048	39.385	35.393	100%
`rho-strong`	online	40	40	34.7	38	0.913	1	0.901	43%

One outlier in the PR binary's `rho-strong` solve (about 3.3 ms against a 0.15 ms median) is in Figure 1 and is kept.

Section 5a band, `ecbench band --dir sessions/fix-claim --ranks` (identical JSON for `base-claim`; the counts do not depend on the binary), S at w = 0, 0.1, 1 GAE per unpriced native operation:

arm	S at 0	S at 0.1	S at 1
ic	16.852	17.283	21.16
rho	1.541	1.635	2.478
rho-strong	4.004	4.14	5.361

Verdict across the band: rho-strong < ic, rho < ic, rho < rho-strong (order holds at every weight; uncounted empty for all three arms). Distance remaining of IC to the matched strong rho on this curve: 4.21, 4.17 and 3.95 times at 0, 0.1 and 1.

Attribution: which change removes the cost

arm (smoke, median ms)	base	fix (trim + probe)	fix-notrim (probe only)	fix-noprobe (trim only)	fix-notrim2 (neither)
rho-neg	5.833	0.25	0.236	0.277	6.57
rho-plain	6.144	0.277	0.25	0.275	6.738
rho-frozen	6.091	0.33	0.281	0.271	6.594
rho-neg-aa	6.767	0.25	0.221	0.278	6.419
bsgs	5.392	0.186	0.188	0.212	6.461
bsgs-il	5.262	0.162	0.14	0.182	5.769
bsgs-neg	5.303	0.137	0.123	0.14	6.177
kangaroo	6.24	0.196	0.172	0.215	8.342
ic-mitm	1.891	1.83	1.866	1.88	1.906

arm (claim, median us)	window	base	fix	fix-notrim	fix-noprobe	fix-notrim2
ic	solve	1412.7	1354.2	1342.1	1456.2	1518.4
ic	online	127.1	119.1	113.5	125.2	120.3
rho	solve	5886.9	249.1	198	296.4	5638.7
rho	online	5885.3	245.6	194.1	292	5635
rho-strong	solve	5658.4	148.7	147.3	170	5631.3
rho-strong	online	34.7	38	37.7	42.6	39.2

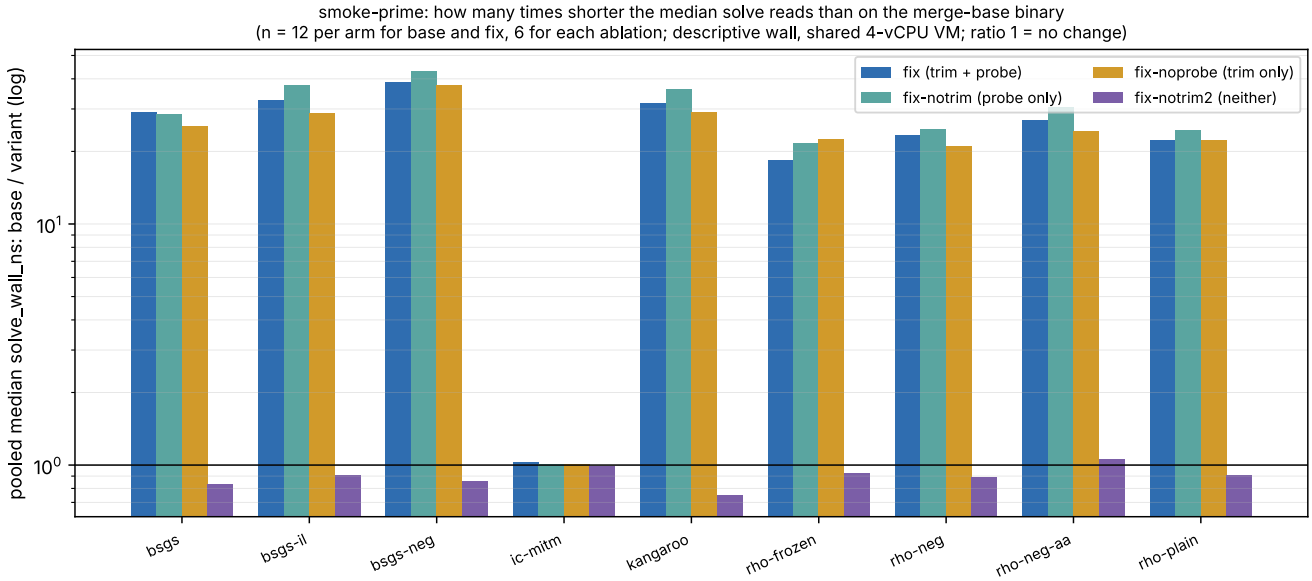


Figure 2: smoke-prime, pooled median solve wall of the merge-base binary divided by each variant's (log scale; 1 = no change). n = 12 per arm for base and fix, 6 for the ablations. Data: analysis/ratios.json.

Read: with the trim or the probe present the PR binary reads 0.12 to 0.33 ms on the eight non-IC smoke arms; with neither it reads 5.8 to 8.3 ms, as the merge-base binary does. The PR's own source comment places the cost in malloc_consolidate of fastbins left by set-up; this study measured that removing both consolidation sources restores the cost, and did not trace malloc_consolidate itself (no tracer on this VM), so the mechanism named is the PR's and the probe's allocation is a candidate explanation for why the trim looked redundant, not an observation.

Verdicts against the preregistration

id	prediction	result
H0 counts identical	every compared field equal	**holds**: 10 pairs, 0 differences
H1 smoke wall	7 of 8 non-IC arms with base/fix >= 2 and fix median < 1 ms	**supported**: 8 of 8 (ratio 18.4 to 38.8; fix 0.137 to 0.330 ms)
H2 claim IC online	base/fix > 1 in each order and >= 70% of fix runs below base median	**inconclusive**: pooled 1.067, orders 1.135 and 0.983, 65% below
H3 audit	every session passes replay-all and the CI form	**holds**: 14 of 14
H4 attribution (exploratory)	notrim near base if the trim causes the drop	**not as predicted**: fix-notrim is at the fix level; AMENDMENT_1: probe also consolidates; with neither the cost returns to base (0.75 to 1.05)
noise floor	A/A ratio per binary	1.160 (base), 1.003 (fix)

Audits (`ecbench verify --replay-all --exit-code`, receipts in `audits/`, not inside sessions):

session	result	session id
`base-claim`	OK: 66 records, 66 verified, 60 replays reproduced, 0 problems	ECBS1h7c0082ac7265
`base-claim-rev`	OK: 66 records, 66 verified, 60 replays reproduced, 0 problems	ECBS1h4d78060e6148
`base-smoke`	OK: 72 records, 72 verified, 54 replays reproduced, 0 problems	ECBS1hd3725d8a32c3
`base-smoke-rev`	OK: 72 records, 72 verified, 54 replays reproduced, 0 problems	ECBS1h420293eaa9c5
`fix-claim`	OK: 66 records, 66 verified, 60 replays reproduced, 0 problems	ECBS1h157a7553e392
`fix-claim-rev`	OK: 66 records, 66 verified, 60 replays reproduced, 0 problems	ECBS1ha418537656ed
`fix-noprobe-claim`	OK: 66 records, 66 verified, 60 replays reproduced, 0 problems	ECBS1h11617fea463f
`fix-noprobe-smoke`	OK: 72 records, 72 verified, 54 replays reproduced, 0 problems	ECBS1h322f7c2815eb
`fix-notrim-claim`	OK: 66 records, 66 verified, 60 replays reproduced, 0 problems	ECBS1hc3e67b6813c4
`fix-notrim-smoke`	OK: 72 records, 72 verified, 54 replays reproduced, 0 problems	ECBS1h628195e31e14
`fix-notrim2-claim`	OK: 66 records, 66 verified, 60 replays reproduced, 0 problems	ECBS1h9d777ef2e237
`fix-notrim2-smoke`	OK: 72 records, 72 verified, 54 replays reproduced, 0 problems	ECBS1h0aa59508bb08
`fix-smoke`	OK: 72 records, 72 verified, 54 replays reproduced, 0 problems	ECBS1hff17ab9636f0
`fix-smoke-rev`	OK: 72 records, 72 verified, 54 replays reproduced, 0 problems	ECBS1h5d29a35e9eba

Scope and limits

- Two sizes (a 20-bit prime-order curve, Koblitz a=1 n=17), one shared 4-vCPU VM below L3, glibc 2.39, one pair of commits. It says nothing about other allocators, larger solves where consolidation is negligible, or L3 hosts.
- `base` and `fix` differ by the whole PR (grading v3, isolation probing, PMU group, session writing); the ablations are scratch builds that do not exist in the repository and change only the lines named in their patches.
- The two binaries ran in different sessions: `ecbench compare` refuses wall time across sessions by design, so the wall comparison is a descriptive summary of recorded values, not an admitted ecbench wall result.
- The ABBA order cancels linear drift only; other agents' builds ran throughout and the cargo lock was held at every session start.
- The PR's "about 15% IC online window" figure is from a session not in this study; this study's claim-koblitz-r10 does not distinguish the two binaries on that window.

Validation gate status

`docs/VALIDATION_GATE.md` requires an independent replay certificate and an independent validator report. Both were produced on the same host by the same agent that ran the sessions, so `independent: true` cannot honestly be asserted. A candidate `VALIDATION.json` with its replay certificate and validator report is in `validation/` with `independent: false`; `scripts/validation_gate.py evidence validation` refuses it on exactly that. No row is added to `benchmarks/README.md`. A replay of `sessions/` from a host of another class, or by a validator who did not run the study, would complete it; the matched-rho reference and the section 5a band (above) are already in place.

Files

`PREREGISTRATION.md`, `AMENDMENT_1.md`, `specs/`, `sessions/` (14, unedited), `audits/`, `comparisons/`, `analysis/` (scripts, `summary.json`, `ratios.json`, patches, tables), `figures/`, `reports/run-*/`, `BINARIES.txt`, `logs/`, `validation/`. Benchmarks copy: `benchmarks/20261023-ecbench-allocator-confound/` (graphs and PDFs, no README row).